ULTRAHIGH RESOLUTION ON-CHIP SEPARATION SYSTEM USING NEGATIVE MAGNETOPHORESIS

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ABSTRACT

A high-resolution magnetic separation microfluidic system used for the negative magnetophoretic separation of nanoparticles is presented in this paper. The chip can generate an ultrahigh gradient magnetic field greater than 10^5 T/m in the separation channel by integrating two on-chip magnetic pole arrays. A label-free separation of nanoparticle is achieved by this microfluidic system with a mixed sample of 0.2 µm and 1 µm polystyrene particles, where the recovery rate of 0.2 µm particles is 91.20%, and the purity reaches 94.72%. The experimental results show that the chip can efficiently separate nano-scale particles with ultrahigh resolution, and in future, it may develop into a versatile and robust tool for the separation and purification of nano-biological samples.

KEYWORDS

Negative magnetophoresis; Microfluidic system; Nanoparticle separation; Ultrahigh gradient magnetic field.

INTRODUCTION

Efficient separation and purification of specific biological samples are the premise and basis for ensuring accurate and reproducible results of subsequent diagnostic and therapeutic research. The rapid development of microfluidic technology provides a powerful tool for the separation and analysis of specific biological sample. Due to the commercial availability with various sizes and surface functional groups of the micro/nano magnetic beads, microfluidic magnetic separation combined with magnetic bead immunoassay is currently the most common biological sample separation method [1]. Compared with other microfluidic separation methods, magnetic separation has the characteristics of no damage to biological samples and a relatively simple and controllable system [2].

Based on the difference in magnetic susceptibility of the biological sample and the surrounding medium, the magnetic separation methods can be classified into magnetophoresis and negative magnetophoresis [3]. Magnetophoresis can precisely control micro/nano-scale magnetic beads, and achieve capture of specific biological samples by binding the magnetic beads to the antigen on the sample of interest [4]. The negative magnetophoresis can be used for label-free sorting of biological samples through magnetic media. Compared with magnetophoresis, this label-free manner makes it more convenient and possesses broad application prospects [5]. Researchers started from the simple use of single or multiple magnets to realize on-chip negative magnetophoretic separation of cells [6-8], to later using of magnet array or combining other forces to continuously improve the separation resolution [9,10], but most literature still stays in separating micrometer-scale cells or particles [11,12].

In this paper, a combination of permanent magnets, high-permeability alloys and a magnetic pole channel was used to build an arbitrary shape magnetic pole array with high magnetic field strength near the separation microchannel (the distance is 3 µm). The magnetic pole array can generate an ultrahigh magnetic field gradient that greater than 10^5 T/m in a specific region of the separation microchannel, which is filled with commercial ferrofluid at an extremely low concentration (0.01%). The separation of nanoparticles is realized in the microfluidic device, demonstrating the potential of this method.

CHIP DESIGN AND FABRICATION

The design of the magnetic device is shown in Figure 1, the chip is consisted of a separation channel, two magnetic pole channels, two rectangle shaped permalloy and two NdFeB magnets. The magnetic pole channel in the separation region is fullfilled with Fe_3O_4 powder.

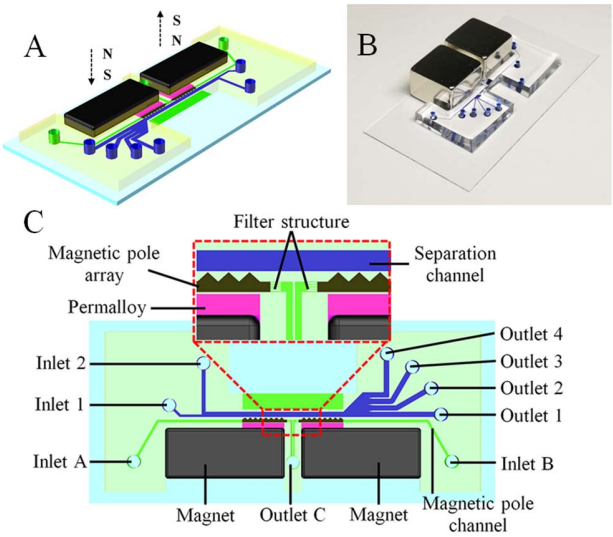

Figure 1: Schematic illustration and photograph of the microfluidic system. (A) Schematic drawing of the chip design; (B) Photograph of prototype chip; (C) Detailed chip design and enlarged view on the separation region.

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The NdFeB magnets can provide a strong external magnetic field, then magnetic field can be transmitted to the magnetic pole channel through the high permeability permalloy. The magnetic pole channels can be designed in any shape and fabricated by soft lithography. In order to generate a high gradient magnetic field, here the magnetic pole array is designed as triangular structures, in which the distance between the top corner of the triangle and the separation channel is 3 μm. Two magnetic pole arrays can be formed near the separation channel with the magnetic pole channels filled with Fe_3O_4 powder. The pole arrays can be magnetized by an externally conducted magnetic field, then a high strength and high gradient magnetic field can be generated inside the separation channel to provide a strong negative magnetophoretic force.

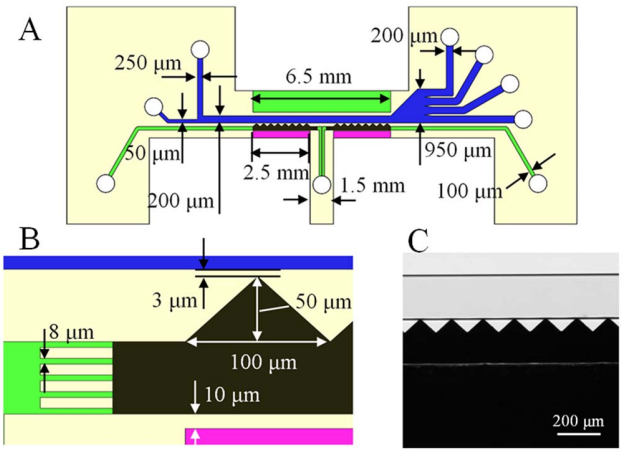

Figure 2: Schematic drawings and photograh of the separation region. (A) Schematic drawing of microchannel size, the height of all channels is 40 μm; (B) Schematic drawing of magnetic pole size; (C) An image of separation region.

Design parameters of the chip and image of the separation area are shown in Figure 2. The microfluidic chip can be fabricated by 6 steps: 1) A PDMS microfluidic layer is fabricated by soft lithography and then bonded to a glass substrate; 2) Rectangle shaped permalloy (1J85) is embedded at the separation channel side (the green region in Figure 2A); 3) NdFeB magnet (N52) is put on the permalloy; 4) Fe_3O_4 powder mixed with pure water (mass to volume m/v = 1:500) is injected into the magnetic pole channel from inlet A and inlet B, then the Fe_3O_4 powder is concentrated in the magnetic pole array area under the action of fluid and permalloy magnetized by the magnet; 5) Liquid PDMS is then injected into the magnetic pole channels, two rectangle shaped permalloy are embedded at the magnetic pole channel side (the pink regions in Figure 2A) to hold the Fe_3O_4 powder, and then the chip is heated by a thermostat at 80°C for 2 hour to cure the magnetic pole array structure; 6) The magnet and the permalloy at the separation channel side are removed, and two NdFeB magnet are put on the permalloy at the magnetic pole channel side.

WORKING MECHANISM

Force Analysis

The nonmagnetic particles immersed in ferrofluid are experienced negative magnetophoretic force $\boldsymbol{F}_{\mathrm{m}}$, hydrodynamic drag force $\boldsymbol{F}_{\mathrm{d}}$ and inertia lift force $\boldsymbol{F}_{\mathrm{L}}$. Magnetic force $\boldsymbol{F}_{\mathrm{m}}$ can cause lateral displacement of particles, which is the key to particle separation, and it can be described by the following equation:

$$\boldsymbol{F}_{\mathrm{m}} = \frac{V_{\mathrm{p}}(\chi_{\mathrm{p}}-\chi_{\mathrm{f}})}{\mu_0}(\boldsymbol{B}\cdot\nabla)\boldsymbol{B} \tag{1}$$

Where V_{p} is the volume of the particle, μ_0 is the permeability of free space, $\boldsymbol{B}$ is the magnetic induction intensity at the location of the particle, χ_{p} and χ_{f} are the magnetic susceptibilities of particle and ferrofluid, respectively.

The trajectory of particles in the microchannel is determined by the above three forces, and the motion control equation can be described by:

$$m_{\mathrm{p}}\frac{\mathrm{d}\boldsymbol{u}_{\mathrm{p}}}{\mathrm{d}t} = \boldsymbol{F}_{\mathrm{m}} + \boldsymbol{F}_{\mathrm{d}} + \boldsymbol{F}_{\mathrm{L}} \tag{2}$$

Where m_{p} and $\boldsymbol{u}_{\mathrm{p}}$ are the mass and the velocity of particles, respectively.

Numerical Simulations

Based on Equation (2), we use COMSOL Multiphysics to simulate the magnetic field distribution and particle trajectory in the separation channel.

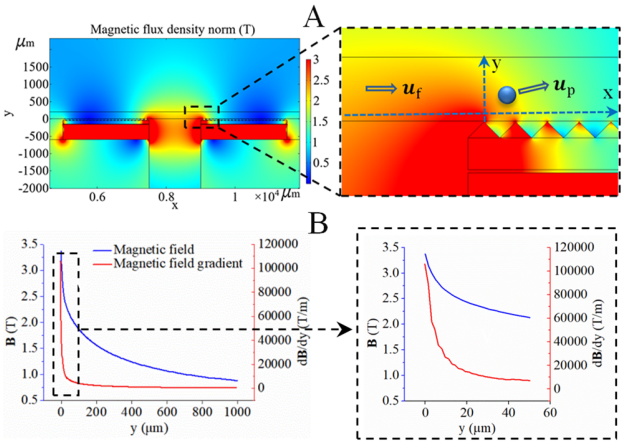

Figure 3: Magnetic field simulation. (A) The magnetic flux density distribution; (B) The magnetic field and its gradient along radial direction (y-direction) at the tip of magnetic pole (x=9050 μm).

The magnetic field distribution is shown in Figure 3.The results show that the magnetic pole arrays generate a high intensity and high gradient magnetic field in the separation channel, especially at both ends of the magnetic pole array. The directions of the magnetic fields of the two magnetic pole arrays are opposite, an ultrahigh gradient magnetic field is formed between the two arrays, and as the distance (the distance along the y-axis in Figure 1) to the tip of the magnetic pole array increases, the magnetic field strength and its gradient decrease rapidly. The highest magnetic field intensity in the separation channel exceeds 3 T, and the maximum magnetic field gradient exceeds 10^5 T/m (near the channel wall). As the distance increases to 30 μm from the channel wall, the magnetic field gradient rapidly decreases to 10^4 T/m. It can be concluded that the particles will experience a larger magnetic force within 30 μm from the microchannel wall (in the side of the magnetic pole array side).

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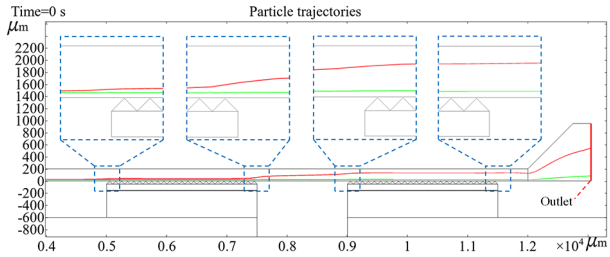

Figure 4: Particle trajectories in the separation channel, green line represents a 0.2 μm particle; red line represents a 1 μm particle.

The particle trajectories are shown in Figure 4. In order to ensure the particles are focused close to the channel wall, the flow ratio of inlet 1 and inlet 2 is set to 1:5, in Figure 4 the flow rate of inlet 1 is set to 0.5 μL/min. The simulation shows that particles of 0.2 μm and 1 μm in size start to deflect laterally (along the y-axis in Figure 1) when they enter the magnetic pole arrays area. Because the magnitude of the magnetic force is different, 1 μm particle and 0.2 μm particle begin to separate. When the particles pass through the region between the two magnetic pole arrays, the magnetic force reaches the maximum, so the two kinds of particles are deflected more obviously here, and the separation distance between the two particles is further increased. The width at outlet of the separation channel is enlarged, on the one hand, it can enlarge the separation distance of the two particles, on the other hand, it can reduce the flow rate, which facilitates subsequent experimental observation. At the outlet of the separation channel, the particles of two sizes have changed from a mixed state to a separated state. We have simulated the particle separation under different flow rates. In next section, we will compare with the experimental results to get the best separation flow rate.

RESULTS AND DISCUSSION

In the experiments, the flow rates are optimized first (the flow ratio of inlet 1 and inlet 2 was kept at 1:5). We use two kinds of polystyrene fluorescent particles with sizes of 1 μm (glacial blue, Bangs Laboratories, Inc.) and 0.2 μm (yellow-green, Thermo Fisher Scientific) in the experiments. Firstly, the 1 μm and 0.2 μm particles are mixed according to a concentration ratio of 1:10 (the concentration of 1 μm particles is ~ 3500 particles/μL, and that of 0.2 μm particles is ~ 35000 particles/μL), here the fluorescent particles are diluted with 0.003X ferrofluid (EMG 805, 3.6%, Ferrotec, Corp.). Then the mixed sample is injected into the chip from inlet 1 using a syringe pump (CETONI NeMESYS, CETONI GmbH), the pure ferrofluid with same concentration (0.003X) is injected into the chip from the inlet 2 as the sheath flow. Finally, the experimental results at the outlet are recorded with a Zeiss Axio Observer 7 inverted fluorescence microscope (Carl Zeiss Microscopy GmbH), and the distribution of the particles of the two sizes at the outlet is analyzed through the fluorescence pictures. The experimental results are compared with the simulation analysis of particle trajectory, as shown in Figure 5.

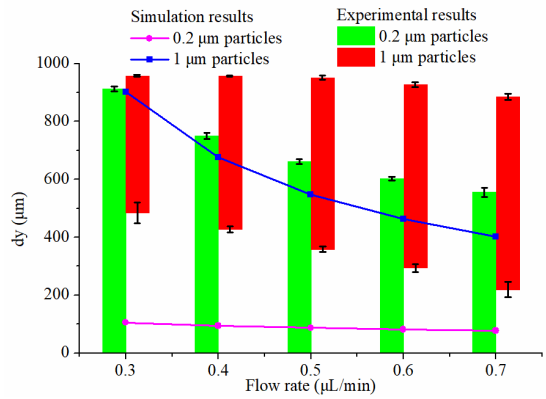

Figure 5: Comparison between simulation results and experimental results on particle distribution along y-axis at the outlet under different flow rates of sample injection.

The simulation results show that as the flow rate (inlet 1) increases, the particle displacement along the y-axis continuously decreases, and the distance between the two types of particles also decreases. Theoretically, the smaller the flow rate, the better the separation of these two kinds of particles, but the sample throughput will also decrease when the flow rate decreases. It can also be concluded from the experimental results that the displacement of the particles along the y-axis decreases with the increase of the flow rate, but the particles have a certain concentration, the trajectory in the experiment is not a straight line, but a distribution within a certain range. Furthermore, there are interactions between large particles and small particles during movement, such as collisions or fluid disturbances, etc. (these are not considered in the simulation), so there is an overlapping area between the two kinds of particles along y-axis direction in the fluorescence images. Therefore, in the experiments, low flow rates may not contribute to separation, because the interaction time of the particles would increase at low flow rates, resulting in more overlap of particles. After counting the length of the overlap area, we obtain the optimal flow rate is 0.5 μL/min (inlet 1), at which the overlap length of the two kinds of particles is the shortest, as shown in Figure 5.

The separation trajectories of the two kinds of particles under the optimal flow are shown in Figure 6A and 6B, in order to increase the contrast, we added a red false color to the blue fluorescent particles (1 μm). Figure 6A shows that 0.2 μm particles are distributed in outlets 1-3, but it can be concluded that the amount of particles at outlet 1 is the most from the fluorescence intensity. Figure 6B shows that the 1 μm particles are distributed in outlets 2-4. In order to accurately and quantitatively analyze the particles at the outlets, we collect the samples from the 4 outlets and then analyze them with a flow cytometer (CytoFLEX S, Backman Coulter). The statistical results are shown in Figures 6C and 6D. The actual concentration ratio of the sample at inlet 1 measured by the flow cytometer is 7.7:1 (0.2 μm particles:1 μm particles), and the overall recovery rate of particles collected from all outlets is 97.36%. The distribution proportions of 0.2 μm particles in outlets 1-4 are 61.79%, 29.41%, 7.71% and 1.09%, respectively. The sum of 0.2 μm particles in outlets 1 and 2 accounts for 91.20% over all outlets (Figure 6C). The proportions of 0.2 μm particles and 1 μm particles in

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outlet 1 are 98.09% and 1.91% (51.36:1), respectively, that is, most of the particles in outlet 1 are the 0.2 μm particles. And the proportions of 0.2 μm particles and 1 μm particles in outlet 2 are 88.35% and 11.65%, respectively (Figure 6D). Taking outlet 1 and outlet 2 as the recovery outlets for 0.2 μm particles, the recovery rate is 91.20%, and the purity reaches to 94.72%, which shows that the chip can efficiently separate and purify 0.2 μm particles from the micro-scale particles.

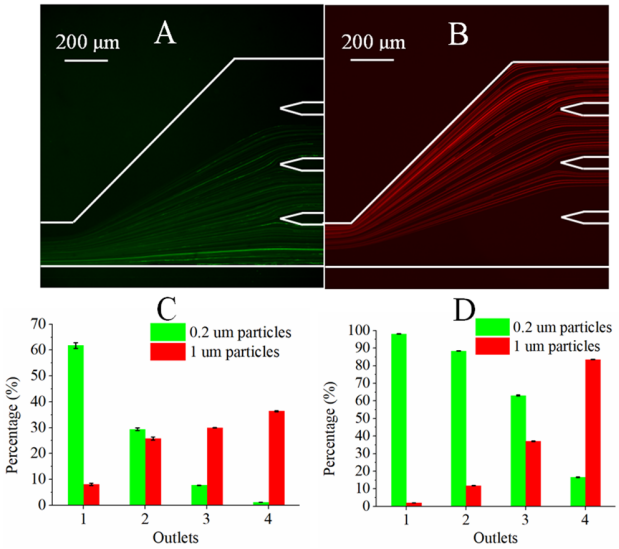

Figure 6: Experimental results. (A) The trajectory of 0.2 μm particles; (B) The trajectory of 1 μm particles; (C) Percentage of particles in each outlet; (D) The proportion of the two kinds of particles in each outlet.

CONCLUSION

In this paper, a negative magnetophoretic microfluidic chip with ultrahigh resolution has been presented. The combination of permanent magnets, permalloy and on-chip magnetic pole arrays generates an ultrahigh gradient magnetic field inside the microchannel, which can separate nanoparticles from 1 μm particles. First, a detailed theoretical and numerical analysis on the magnetic field and particle trajectories is carried out, the magnetic field gradient inside the separation channel was greater than 105 T/m. Then the experiments on a mixed sample of 1 μm and 0.2 μm particles in ferrofluid with an extremely low concentration (0.01%) verify the performance of the chip. The experimental results show that the overall recovery rate of particles is 97.36%, the recovery rate of 0.2 μm particles is 91.20%, and its purity reached 94.72%. The proposed microfluidic device will not only boost the field of negative magnetophoresis for sample separation and purification, but also open new opportunities for developing new methods for research and application of nanoscale biological samples.

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REFERENCES

[1] H. Xu, C. Liao, P. Zuo, Z. Liu, B. Ye, "Magnetic-based microfluidic device for on-chip isolation and detection of tumor-derived exosomes", *Anal. Chem.*, vol. 90, pp. 13451-13458, 2018.

[2] Q. Cao, X. Han, L. Li, "Configurations and control of magnetic fields for manipulating magnetic particles in microfluidic applications: magnet systems and manipulation mechanisms", *Lab Chip*, vol. 14, pp. 2762-2777, 2014.

[3] S. Yan, J. Zhang, D. Yuan, Q. Zhao, J. Ma, W. Li, "High-throughput, sheathless, magnetophoretic separation of magnetic and non-magnetic particles with a groove-based channel", *Appl. Phys. Lett.*, vol. 109, no. 214101, 2016.

[4] Y. Wan, G. Cheng, X. Liu ,S. Hao, M. Nisic, C. Zhu, Y. Xia, W. Li, Z. Wang, W. Zhang, S.Rice, A. Sebastian, I. Albert, C. Belani, S. Zheng, "Rapid magnetic isolation of extracellular vesicles via lipid-based nanoprobes" , *Nat. Biomed. Eng.*, vol. 1 , pp. 1-11, 2017.

[5] A. Kose, B. Fischer, L. Mao, H. Koser, "Label-free cellular manipulation and sorting via biocompatible ferrofluids", *Proc. Natl. Acad. Sci.*, vol. 106, pp. 21478-21483, 2009.

[6] T. Zhu, R. Cheng, S. Lee, E. Rajaraman, M. Eiteman, T. Querec, E. Unger, L. Mao, "Continuous-flow ferrohydrodynamic sorting of particles and cells in microfluidic devices", *Microfluid. Nanofluid.*, vol. 13, pp. 645-654, 2012.

[7] L. Liang, C. Zhang, X. Xuan, "Enhanced separation of magnetic and diamagnetic particles in a dilute ferrofluid", *Appl. Phys. Lett.*, vol. 102, no. 234101, 2013.

[8] W. Zhao, R. Cheng, S. Lim, J. Miller, W. Zhang, W. Tang, J. Xie, L. Mao, "Biocompatible and label-free separation of cancer cells from cell culture lines from white blood cells in ferrofluids", *Lab Chip*, vol. 17, pp. 2243-2255, 2017.

[9] A. Munaz, M. Shiddiky, N. Nguyen, "Magnetophoretic separation of diamagnetic particles through parallel ferrofluid streams", *Sens. Actuators, B*, vol. 275, pp. 459-469, 2018.

[10] A. Shamloo, M. Besanjideh, "Investigation of a novel microfluidic device for label-free ferrohydrodynamic cell separation on a rotating Disk", *IEEE Trans. Biomed. Eng.*, vol. 67, pp. 372-378, 2019 .

[11] W. Zhao, T. Zhu, R. Cheng, Y. Liu, J. He, H. Qiu, L. Wang, T. Nagy, T. Querec, E. Unger, L. Mao, "Label□free and continuous□flow ferrohydrodynamic separation of HeLa cells and blood cells in biocompatible ferrofluids", *Adv. Funct. Mater.*, vol. 26, pp. 3990-3998, 2016.

[12] H. Kye, B. Park, J. Lee, M. Song, H. Song, C. Ahrberg, B. Chung, "Dual-neodymium magnet-based microfluidic separation device", *Sci. Rep.*, vol. 9, pp. 1-10, 2019 .

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